

Team 3

Amber Sun, Daniel Park, Lincoln Williams, Rachel Lee, Ruth Kim

Pipe Traversing and Relining Worm Robot

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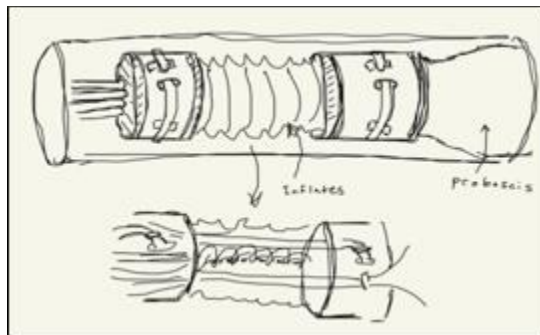


Figure 1. Final Design with Improved Inflating Sections and Air Pump System [1].

Introduction and Problem Description:

Pipes deteriorate over time and can contaminate a water system due to corrosion, sediment buildup, water age, and biofilm development [2]. In a study conducted by Rogers and Grigg, the aging water pipeline infrastructure in the United States experiences approximately 30 breaks per 100 miles of pipe per year [3]. However, replacing deteriorated or contaminated pipes is expensive. The American Water Works Association estimates repair and maintenance costs totaling to approximately one trillion dollars in the next 25 years alone [4]. The continued use of deteriorated and contaminated pipes increases the risk of disease transmission, especially in urban centers, and prevents people from accessing vital water resources.

As the United States' pipe infrastructure ages, the cost and labor required to inspect deteriorating pipes and maintain them will continue to grow. To address the growing need for an efficient and automatic method of inspecting pipes, a robot will be developed to remotely inspect and stress test pipes to identify areas of deterioration. Currently, as per federal regulations, water pipes undergo hydrostatic tests, or pressure tests, before use and only periodically after installation. During a pressure test, a pipe is subjected to stress up to 90% of its yield strength to ensure its integrity [5]. However, after installation, stress testing is done less frequently, leaving a critical gap in monitoring the structural health of the pipes [6]. By making pressure testing feasible during use and not limited to solely before the use of a pipe, the robot will provide a targeted approach to diagnosing deteriorating pipes for replacement. The robot will be able to navigate through pipes to perform strength tests and collect quality data, improving early detection of deteriorating pipes, making repairing pipes a more efficient and targeted process, and ultimately safeguarding communities from the consequences of pipe deterioration.

Design and Build:

Prior to building or designing our robot, our team began by selecting success metrics and project goals. We decided that our robot should be able to fit inside and navigate through pipes ranging from 6 to 16 inches in diameter, the standard range of main-line water pipes [7]. Similarly, our robot should be able to bend at least 90 degrees in turns of radius of 6 inches to follow the potential curvature of standard water piping, again ensuring that the robot can traverse the pipes themselves. Our robot should also be able to reach speeds of up to 20 feet per hour minimum as most main water pipes are a minimum of 10 feet, to minimize downtime and make our robot feasible and practical in field use [8]. Additionally, this speed requirement was imposed to ensure that pipe access will not be disrupted for an overly extended period of time, as the pipe will have to be out of service during inspection. The robot itself should weigh less than 20 pounds to again allow for ease of use and have a battery life of at least two hours, again ensuring ease of use and feasibility. The robot itself should also be waterproof and have minimized on-board electrical components, ensuring no issues occur when in contact with water or pipe residue. Mitigating electricity then became one of our guiding principles, as we later homed in on a purely air-powered system. After we settled on these considerations and principles, we began developing the two main mechanisms of the robot: the re-liner and the movement system.

The first design system we considered was the pipe re-liner. Our team was initially inspired by current pipe-relining systems, from which the instillation process resembles the ribbon worm's proboscis defense tactic.



Figure 1. Ribbon worm with extended proboscis [9].

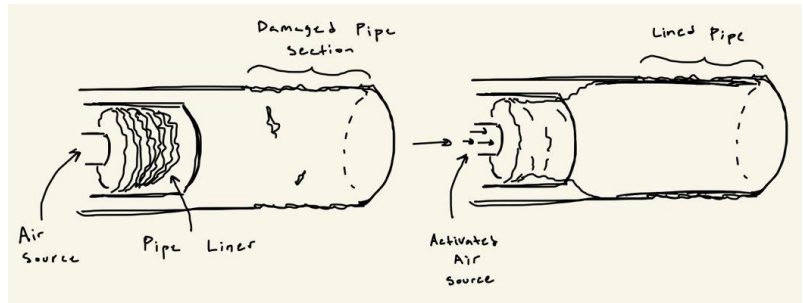


Figure 2. Robotic proboscis sketch [10].

The ribbon worm then became our primary bio-inspiration. We felt implementing a similar but more precise system would then allow for greater efficiency and cost-effectiveness, which then became our first design challenge. We sought to unfurl the re-liner on-board our robot via air power, but delivering the air to the proboscis became our first design challenge. We could either place an on-board air compressor or run an air tube out of the back of the robot to feed the air outside of the pipe. The on-board air compressor would then create the challenges of forming a battery system and a potential remote control, responding to changes in air pressure inside the pipe, and securely housing the electrical components within the robot as to prevent electric contact with any water residue from the pipe. Contrarily, the tube-fed system would omit these issues, with the only drawback being the requirement of a long air tube spool for pipe exploration. However, we quickly realized that this tube-spool could then serve as a recovery system if our robot were to break down, as it would then act as a direct line which could be pulled for manual extraction. In turn, the tube-fed system only provided benefits and became our first design choice.

When considering our movement system, we realized that the system should be considered in two respects: the level of bioinspiration and the power system. When first naming potential movement styles, two main options became apparent: rolling or inching along the pipe. Rolling would be centered around a wheel or track system, while inching would be based around inflating sections or radially expanding robotics. In our discussions, we quickly eliminated the radial expansion option for the inch-worm system, as such technology is relatively complex and would require very precise 3D-printed or handmade parts. Additionally, such technology has many moving parts and would likely require electricity and extensive wiring, therefore again creating the problem of an on-board battery system. After this first elimination, we sketched our first three movement prototypes (centered around tracks, wheels, and inflating sections) and quickly made more design decisions based on feasibility and our technical skills. After visiting 725, we formed three bare-bones prototypes demonstrating the potential movement systems. These prototypes and sketches prompted us to eliminate the wheel and track systems, as they similarly require electricity, lessen the overall bioinspiration of the robot, and are also much less adjustable to different pipe widths. Additionally, the bare-bones track and wheel prototypes were far more difficult to assemble with far more moving parts than the inflating sections. The

inflating sections option then became the focus of our efforts, as it would have the benefits of only requiring air power (eliminating the problem of electricity and insulation against any water during movement) and closely mirrors the ribbon worm's natural movement system.

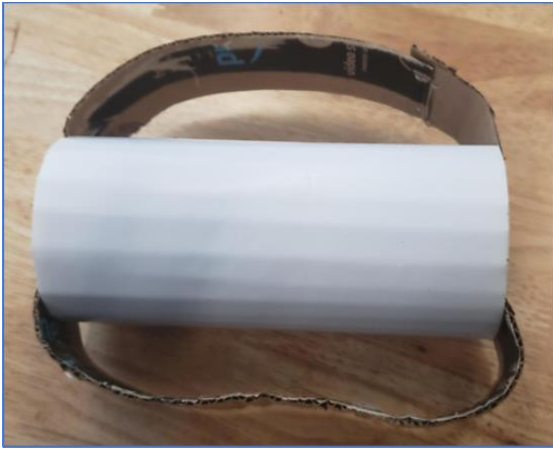


Figure 2. First track-movement prototype [11].

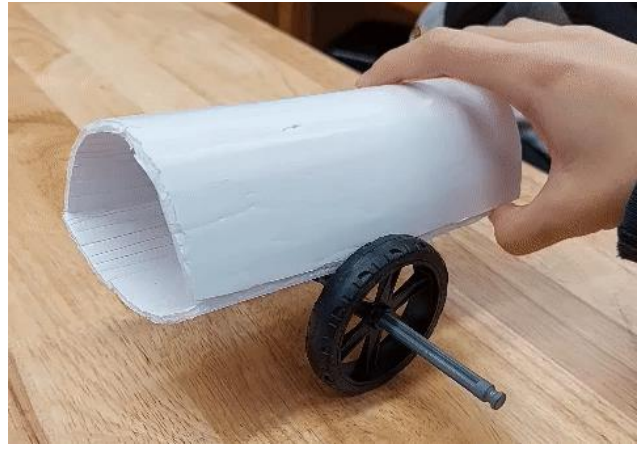


Figure 3. First wheel-movement prototype [12].



Figure 4. First sectional-movement prototype [13].

Our first prototype then acted as a combination of the proboscis system and the inflating sections movement system. The body was formed from two rigid cardboard tube sections that were each wrapped in latex balloons, acting as the gripping sections of the movement system. The cardboard sections were then connected via tubes and a third balloon, forming the center inflation step of the movement system. At the front of our robot, we secured a final balloon to the frontal opening of the leading cardboard tube, forming the proboscis re-liner system. Each balloon of the robot was then attached to a flexible tube to be powered by a hand air pump.



Figure 5. First completed worm-robot prototype [14].

After testing, the balloon-proboscis and side-inflating sectional systems worked as intended. However, the middle inflating section needed improvement, as it would expand outwards towards the pipe walls rather than pushing the cardboard sections apart. In turn, we invested in long animal balloons, as they expand primarily in one direction before expanding outward. Additionally, testing proved that delivering air by hand-pumps was very difficult and physically taxing, leading us to invest in three electronic auto-inflating air pumps for each aspect of the sectional movement system. With these results, we formed our second and final prototype, focusing primarily on improving the locomotion, balloons, and air-deliverance system of the inflating sections (the proboscis had worked as intended, so we decided to omit it in the final design).

Building the final design closely mirrored the process of our prototype, except for replacing the latex balloons loons used in the inflating sectional system with animal balloons. We also connected the two cardboard sections via a folded paper cover that would ensure that the center animal balloon would correctly inflate forwards to propel the robot. Additionally, we drilled holes in the cardboard and fed the flexible air tubes through the robot, in turn creating a sleek design with no exposed tubes (the tubes trailed out of the robot's backside, as intended in the original planning). These tubes were then connected to the automatic pumps, forming an ergonomic air-power system as intended. This concluded our design and build process.

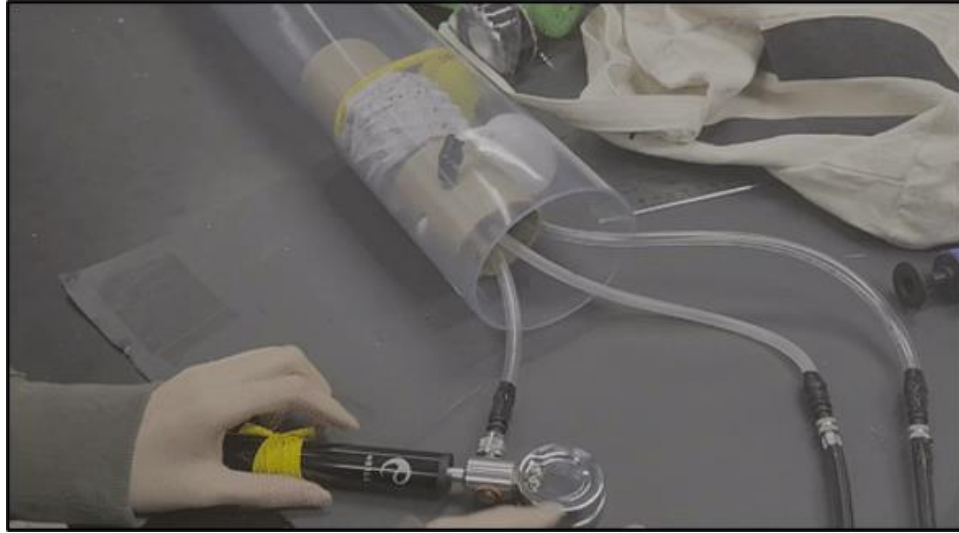


Figure 6. Completed worm-robot within pipe with sectionals inflated [15].

Testing and Results:

For testing, our team primarily looked at the effectiveness of the air-powered movement system. Our final design utilized a balloon as the inflatable material. Because the balloon had good frictional properties (so it did not slip off the walls easily), and was also able to be quickly inflated, the material was outstanding in comparison to other options such as fabric or paper.

To determine how we should inflate the balloon, we tested how much air pressure the balloon would withstand before popping (in psi). The results averaged at 3 psi. As a result, during every inflation of the balloons, we made an effort to stop inflating once the internal psi was around 2 to 2.5 psi.

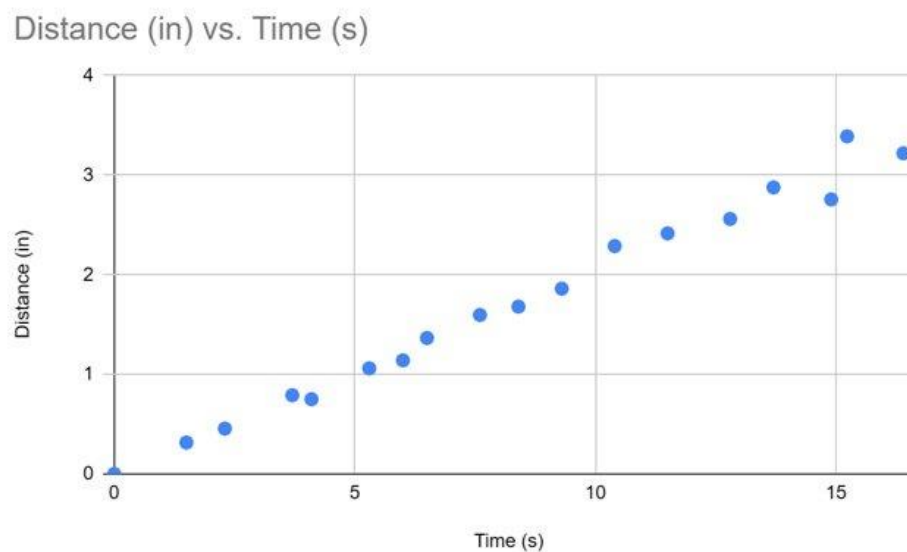


Figure 7. Graph depicting distance versus time data [16].

The balloon took an average of 8 seconds to inflate completely from a deflated state, and an average of 8 seconds to deflate once the valve was opened. Using this data, we calculated that since the inflation of parts is sequential, it would take 48 seconds for one full rotation of the mechanism. The length of the balloon spanned 3.9 inches when deflated, and 1.5 feet when inflated to just below 3 psi. This resulted in a 1.15 feet gain in distance, and an uncertainty of approximately 2 inches due to external impacts such as friction or rubber slipping. However, extrapolating the data measured from a short distance gave us an average of 47 seconds per foot. Since the design specification was to have the robot be able to move at least 20 feet an hour, this has been well reached with a maximum speed of 79 feet per hour.

The balloon, although it was an elastic material, went through deformation after continued use, and slowly was able to take more air pressure before popping. After 20 inflations, 3.5 psi was not enough to pop the balloon anymore, so results varied slightly, as the rate of inflation was also impacted by the change. Because the amount of pressure required to inflate the balloon was reduced, the final length of the center was also impacted, going from 1.5 feet to 1.6 feet, a 6.66% increase, which is non-negligible.

Conclusion

The guidelines that were set at the start were met, as the machine was able to travel almost 4 times the requirement, at 79 feet an hour. In addition, the balloons allowed for the machine to fit different diameters of pipes with different levels of inflation, which could be controlled by adjusting the maximum psi. The two segments of the machine, connected with a loose mid-section, allowed for flexible travel. The maximum angle of bend being 210 degrees surpassed the requirement of being able to turn at least 90 degrees. The machine weighed less than 20 pounds, at a light 1.6 pounds including the 3 air pumps.

Although the main body was successfully made without any electronic parts, it was not waterproof due to the cardboard tubing and paper connector used. The main goal was to test the movement of the machine, which is why this was not an issue during the building and testing. In addition, most of the materials were gathered from the local Makerspace lab, which only offered paper tubes of one size. However, there is still a call for future improvement as having a waterproof body is essential for travelling through wet pipes.

With these enhanced abilities of the machine, the original issue of being unable to fix pipes can efficiently be solved. When there are cracks in pipes, the entire pipe no longer has to be re-lined, wasting several feet of material. Rather, the new pipe-lining machine can travel to the site of damage and re-line that particular section. This not only saves money but also saves resources that could be used for other damaged sections or pipes. Additionally, the pipe machine can be useful in several ways other than simply re-lining. For example, if a camera is attached to the front of the robot and sent to traverse the pipe, inspectors will be allowed to view the interior and potentially diagnose causes of pipe deterioration such as sediment buildup or rust. The machine can also be modified to be able to clear and push out any foreign objects and buildup that hinder the pipe's water flow.

Since the final prototype confirmed the feasibility of using an air-powered movement mechanism, in future iterations the movement mechanism can be streamlined and automated. In

the final prototype, three separate air tubes each connected to separate air pumps were used to inflate the three air balloons needed for movement. While this setup fulfilled its purpose as a proof-of-concept, it resulted in an added complexity to the operation of the robot, as the operator needed to know the sequence of air pumps to use to inflate the correct balloons in the correct order so that the robot would move and to what psi the balloons could be inflated to without popping. In future iterations of the robot, the process of operating the robot will be simplified so that more people can easily use the robot. While more research will have to be done to determine exactly how the movement mechanism can be automated, replacing the separate air tubes with one tube and using automatic valves that can direct the flow of air to a specific balloon may be a potential method. Furthermore, in the future, research will have to be done into what materials the robot should be made of to improve its durability and ensure it is waterproof.

Throughout the process of brainstorming, designing, and developing our robot, we considered the ethics surrounding fair access to clean drinking water. As water is a vital resource essential for everyday life, it is of utmost ethical importance that communities can reliably access clean water. However, socioeconomic factors restrict the amount of pipe infrastructure within some communities, making it even more important that there is proper maintenance in these areas with fewer pipes [17]. Pipe downtimes and repairs must be done quickly, reliably, and affordably. However, current 3rd-party pipe sealant and repair solutions are expensive and sometimes require construction to access pipes, further lengthening repair times and costs. For example, the cost to re-line a single-family home can be up to \$5500 [18], with most insurance companies not covering repairs to pipe and sewer lines [19]. These factors can place such solutions out of the feasible price or timing range of some families and communities. Our solution should then consider the time and cost of relining pipes, ensuring that it can be affordably accessed in all communities regardless of socioeconomic standing and in a timely manner. To minimize costs, our solution should minimize construction, a standard step that greatly lengthens downtime and heightens infrastructure repair costs; our solution should also be installed by a small team of people to further lower costs.

Furthermore, new ethical considerations arose when deciding on future use and improvements of the robot. We considered possible unintended consequences of our solution. If our solution acts as an optimized form of pipe repair, then companies could take advantage of our design, make similar iterations, and hike up the installation price. Moreover, if our solution becomes the industry standard, other pipe-lining solutions could become obsolete, thus harming other businesses specializing in similar forms of pipe repair. As we perfect and start implementing our robot, we should then be conscientious of our competitors to ensure fair use and prevent a potential pipe-sealant monopoly. In addition, as we discussed in the socioeconomic considerations, we must avoid biases in economic class and quality of infrastructure. For example, some communities' pipes may be much older than others and thus have different parameters (like pipe widths or materials, for example) than newer/freshly installed pipes. To then avoid basing our robot on a specific communities' pipe specifications, we decided to make our robot able to traverse different pipe widths and materials as discussed in the introduction and proposed robot specifications. Moreover, as we hope to improve the material that the robot is built from to improve the robot's sturdiness and to ensure that the robot is waterproof, we must also consider potential health concerns associated with certain materials. As the robot will be traversing through water pipes, there will also exist a risk of the robot contaminating the water supply. To ensure that the quality of the water will not be harmed, the materials that the robot

will be made of must be evaluated for toxicity, risk of degradation, and antimicrobial properties. As the worm-inspired pipe inspection robot is improved and developed, by approaching these ethical considerations and those that will arise in the future with the intention of helping the maximum number of people get reliable access to clean water, the way we monitor and maintain critical infrastructure will be revolutionized into a more efficient, cost-saving, and proactive process.

Process Reflection

Throughout this project, our team put forth efforts to design a worm-inspired locomotion robot that aims to inspect for pipe deterioration. The probable impacts of our design prompted us to adopt a broader perspective and evaluate the project from a more comprehensive viewpoint. The decisions we made were based off a deeper understanding of design implementations and complexity.

As we started to brainstorm approaches for the prototype, we discovered factors we had not previously considered through testing and proposing ideas. This gave us space and allowed us to think critically about our design at every angle of the problem, refining what we know to be achievable or not achievable. It helped us recognize the important aspects of our design we needed to focus on. A crucial part of that teamwork dynamic came from our open communication. Building ideas of each other and constantly checking in greatly contributed to the team's overall smooth progress.

One such example is illustrated when we discussed our movement system, how the robot would navigate the pipe or testing environment. We had two main ideas, one being a spring mechanism and another being air pressure. While both seemed feasible, we did not consider the intricate details of how it would work such as how air pressure could be achieved and integrated into our design. As we devised a solution for bringing air flow into the system, the problem of how air flow would move out arose and caused conflict with the design's physical constraints. The more considerations we uncovered, the more complex we realized our design would be. However, it did not deter us as our continuous deliberation and problem solving eventually produced a solution that we all saw potential in.

We learned at the end of our sessions that while we may seek our design in its most ideal state, we needed to be realistic and focus on a design with a better chance of success. The entire engineering design process taught us that risks should be taken, but we need to be smart when taking them. What we did well overall was the team dynamic and how that positively influenced our smooth design process, everything was in order, and we mostly knew what to do. We believe that our communication and problem-solving skills are what we will take with us for the rest of our engineering career. This is an aspect that we realized made our work easier, both more efficient and effective.

Thinking ahead, we would look forward to making future improvements to our design if we had more time. Some developments we could make are utilizing higher-grade materials after we refine and produce the best version. This would increase the quality and make the project more fail-proof as it develops into a tool many industries can use. Another consideration is incorporating a spring mechanism to make it a more streamlined single system.

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